Kennedy Baptist College

Semester One Examination, 2017

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section Two: Calculator-assumed

Your name ____

Your teacher (please circle) Hill Wallace Scorer Foxton

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor
This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	51	35
Section Two: Calculator-assumed	14	14	100	100	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet.

3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.

4. Additional working space pages at the end of this Question/Answer booklet are for planning or continuing an answer. If you use these pages, indicate at the original answer, the page number it is planned/continued on and write the question number being planned/continued on the additional working space page.

5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

6. It is recommended that you do not use pencil, except in diagrams.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed 65% (100 Marks)

This section has **fourteen (14)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 8 **(4 marks)**

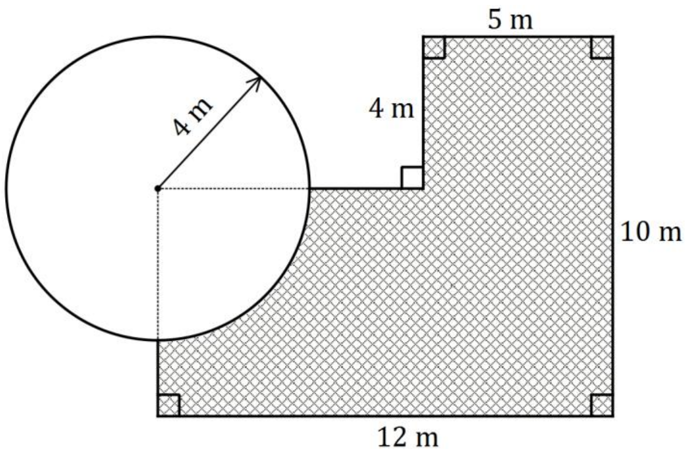
An aunt plans to deposit $7 500 into a savings account for five years, so that when her nephew turns 17 she can help him buy a car.

Her bank has two accounts that she is considering: a 'Basic' savings account that is offering simple interest of 4.2% per annum; and a 'Builder' savings account that is offering an interest rate of 3.9% per annum, compounded annually.

State, with justification, which account you would recommend the aunt uses.

Question 9 **(7 marks)**

The diagram below shows the dimensions of a paved area (shaded) that is adjacent to a circular pool of radius 4 m.



(a) Determine the length of the perimeter of the paved area. (3 marks)

(b) Determine the area of the paving. (4 marks)

Question 10 **(6 marks)**

A tradesman uses the spreadsheet below to quote customers the price of a job ($\$P$), depending on the number of days the job will take (D) and how many labourers (L) will be required. The prices all include GST at 10%.

The formula used in the spreadsheet is $P = 220D + 85L + 150$.

Job prices ($\$P$)		Length of job in days (D)			
		1	2	3	4
Number of labourers (L)	1	455	675	A	1115
	2	540	760	980	1200
	3	625	B	1065	1285

(a) For a job lasting four days and requiring three labourers:

(i) state the price the tradesman will quote. (1 mark)

(ii) calculate how much GST is included in the quote. (1 mark)

(b) Determine the values of A and B in the spreadsheet. (2 marks)

(c) The tradesman quoted a price of $1 895 for a job lasting 6 days. Determine the number of labourers required for this job. (2 marks)

Question 11 **(9 marks)**

The Disability Support Pension payments per fortnight for qualifying persons under 21 years of age who are single with no children are shown in the table below.

Status	Payment
Single, under 18 years of age, at home	$364.20
Single, under 18 years of age, independent	$562.20
Single, 18-20 years of age, at home	$412.80
Single, 18-20 years of age, independent	$562.20

Persons who receive this pension can earn up to $164 fortnightly with no reduction in payment. Earnings over this figure will reduce the payment by 50 cents for each dollar over $164.

(a) A 19-year-old person who lives at home qualifies for this pension and has a part time job for 24 hours per fortnight that pays $16.50 per hour.

(i) Calculate the fortnightly earnings from their job. (1 mark)

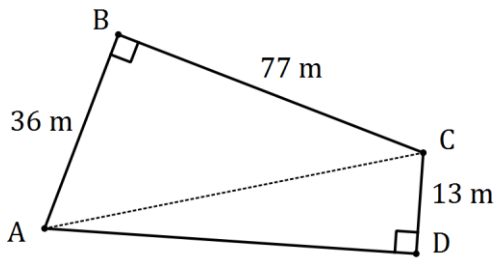
(ii) Determine by how much, if at all, their fortnightly pension will be reduced. (2 marks)

(iii) Determine the fortnightly sum of earnings and pension for this person. (2 marks)

(b) An 18-year-old person who lives independently qualifies for this pension and is also paid \$17.75 an hour for 20 hours work each week. Determine the fortnightly sum of earnings and pension for this person. (4 marks)

Question 12 **(8 marks)**

A playground has the shape of quadrilateral $ABCD$ shown below, where the lengths of AB, BC and CD are 36 m, 77 m and 13 m respectively. Angles B and D are both $90°$.



(a) Determine the lengths of

(i) diagonal AC. (2 marks)

(ii) side AD. (2 marks)

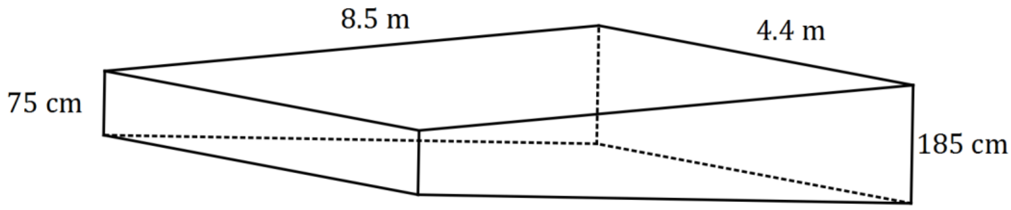
A square playground $PQRS$ has the same area as the playground $ABCD$.

(b) Determine the area of playground $ABCD$. (2 marks)

(c) Determine the perimeter of the square playground $PQRS$. (2 marks)

Question 13 **(6 marks)**

The depth of water in a rectangular swimming pool increases steadily from 75 cm at the shallow end to 185 cm at the deep end. The pool has vertical walls and is 4.4 m wide and 8.5 m long.



(a) A blanket to cover the water surface is to be made from material costing \$6.60 per square metre. Determine the cost of material for the blanket. (2 marks)

(b) Determine the capacity, in kilolitres, of water in the pool. (4 marks)

Question 14 **(9 marks)**

The table below shows the budget used by a young person for running their medium-sized car last year.

Item	Average Weekly Amount
Depreciation	\$72.50
Loan Repayment	\$55.25
Registration	\$7.75
Insurance	\$23.40
Fuel	\$18.30
Tyres	\$2.65
Servicing	\$21.90

(a) Determine the

(i) weekly total of these amounts. (1 mark)

(ii) the average **monthly** amount required to run the car, assuming there were exactly 52 weeks last year. (2 marks)

For this year, the young person expects the average weekly amount for depreciation to fall by 10%, the loan repayment to remain unchanged and all other amounts to increase by 5%.

(b) Determine the new average weekly amounts for

(i) depreciation. (1 mark)

(ii) insurance. (1 mark)

(c) Calculate the new total weekly amount required to run the car this year. (2 marks)

(d) Determine the percentage change in the total weekly amount to run the car from last year to this year, correct to two decimal places. (2 marks)

Question 15 **(8 marks)**

A useful measurement for a photographer when setting up their digital camera for a picture is the hyperfocal distance, D. The photographer knows that every object at a distance greater than half of D from the camera will be in focus.

$$D = \frac{f^2}{Nc} + f$$

In the formula above for the hyperfocal distance D, f is the focal length of the lens, N is the aperture number and c is the diameter of the circle of confusion. All measurements must be in millimetres, apart from the aperture number.

(a) Calculate D when

(i) f is 35 mm, N is 8 and c is 0.019 mm. (2 marks)

(ii) the diameter of the circle of confusion is 0.023 mm, the focal length of the lens is 135 mm and the aperture number is 5.6, giving your answer rounded to the nearest metre. (3 marks)

(b) A photographer is using a lens with a focal length of 35 mm and a circle of confusion of diameter 0.03 mm. What aperture number should the photographer choose from the eight available (2.8, 4, 5.6, 8, 11, 16, 22 or 32) to ensure that all objects greater than 1.3 metres from the camera are in focus? Justify your answer. (3 marks)

Question 16 **(3 marks)**

An inflatable swimming pool full of water is punctured and slowly begins to leak. The volume of water in the swimming pool (V) in litres, t hours after it is punctured is given by the formula

$$V = 6000 - 7.5t$$

a) What is the initial volume of water in the swimming pool? (1 mark)

b) What is the volume of water remaining in the swimming pool after 120 hours? (1 mark)

c) How long will it take for the volume of water in the swimming pool to be half of what it was initially? (1 mark)

Question 17 **(4 marks)**

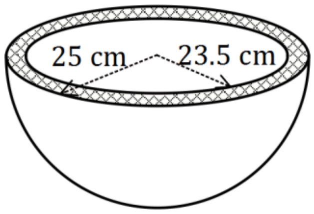
A junior football club conducts a raffle. In total, 350 tickets are expected to be sold, at \$3 for members and \$5 for non-members. The club requires the money raised from ticket sales to cover the cost of the prize, which was \$500, as well as new uniforms which cost \$830.

a) Assuming the football club is to exactly achieve its goal and its expectation regarding ticket sales is correct, write an equation which can be solved to find the number of tickets sold to members. (2 marks)

b) State the number of tickets sold to members and the number of tickets sold to non-members. (2 marks)

Question 18 **(9 marks)**

A hollow plastic hemisphere of external radius 25 cm and internal radius of 23.5 cm is shown below.



(a) Show that the internal surface area of the hemisphere is close to 3 500 cm^2. (2 marks)

(b) Calculate the area of the circular cross-section, shaded in the diagram. (2 marks)

(c) Determine the volume of plastic in the hemisphere. (3 marks)

(d) A solid cube of plastic measuring 75cm by 75cm by 75cm is to be melted and fed into a moulding machine that makes the hollow hemispheres. If there is no wastage, how many hollow hemispheres can be made? (2 marks)

Question 19 **(9 marks)**

The pay rates for full and part time 17-year-old hotel employees are shown in this table.

Hourly pay rate	Evening - Monday to Friday - 7pm to midnight	Night work - Monday to Friday - midnight to 7am	Saturday	Sunday
\$10.93	\$10.93 per hour plus \$2.06 per hour or part of an hour	\$10.93 per hour plus \$3.09 per hour or part of an hour	\$13.66	\$19.13

Meal and other breaks taken during a working day are unpaid.

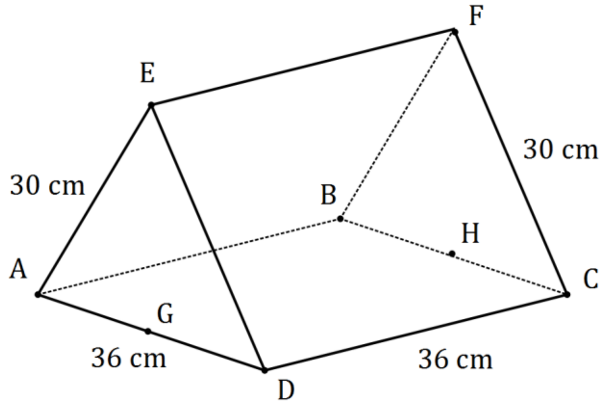
(a) Calculate the weekly wage for a 17-year-old who worked from 8 am until 3 pm from Tuesday to Saturday inclusive, taking a break from 11 am until 11.30 am each day. (3 marks)

(b) A 17-year-old was asked to cover for an absent staff member from 3 pm to 11 pm one Thursday, taking a meal break from 7.15 pm to 8 pm. How much did they earn this day? (3 marks)

(c) A 17-year-old earned \$360.69 for working the same daily hours for five consecutive days, from Friday through to Tuesday inclusive. If they started work at 9.30 am and took a lunch break from 12.30 pm to 1.15 pm, what time did they finish work each day? (3 marks)

Question 20 **(10 marks)**

A triangular prism $ABCDEF$ has a square base $ABCD$ of side 36 cm, two triangular ends and two sloping rectangular faces measuring 36 cm by 30 cm, as shown in the diagram below. G and H are the midpoints of sides AD and BC respectively.



(a) Calculate the length of the diagonal FD in rectangle $CDEF$. (2 marks)

(b) Determine the length EG. (2 marks)

(c) Calculate the area of triangle AED. (2 marks)

(d) Determine the total surface area of the triangular prism. (2 marks)

(e) Calculate the volume of the triangular prism. (2 marks)

Question 21 **(8 marks)**

A supermarket stocked three brands of paper towels, as shown in the table below.

Brand name	Absorb	Blotter	Spongy
Price per roll	$1.95	$2.65	$2.50
Number of sheets per roll	160	220	200

(a) Use the unit cost method to

(i) calculate the price per sheet of each brand. (2 marks)

(ii) state which brand is cheapest per sheet and state the price per 100 sheets of this brand. (2 marks)

A consumer watchdog noticed that the size of paper sheets varied between brands, as shown in this table.

Brand name	Absorb	Blotter	Spongy
Size of one sheet from roll (cm)	21×21	18×21	20×21

(b) Use the unit cost method to calculate the price per square centimetre of paper towel for each brand and comment on whether these new measures support your answer to (a)(ii). (4 marks)

End of questions

Additional working space

Question number: _________